

```
④ # Pause.  
sleep(0.5)
```

The new method `_ship_hit()` coordinates the response when an alien hits a ship. Inside `_ship_hit()`, the number of ships left is reduced by 1 ❶, after which we empty the groups bullets and aliens ❷.

Next, we create a new fleet and center the ship ❸. (We'll add the method `center_ship()` to `Ship` in a moment.) Then we add a pause after the updates have been made to all the game elements but before any changes have been drawn to the screen, so the player can see that their ship has been hit ❹. The `sleep()` call pauses program execution for half a second, long enough for the player to see that the alien has hit the ship. When the `sleep()` function ends, code execution moves on to the `_update_screen()` method, which draws the new fleet to the screen.

In `_update_aliens()`, we replace the `print()` call with a call to `_ship_hit()` when an alien hits the ship:

```
alien_invasion.py def _update_aliens(self):  
    --snip--  
    if pygame.sprite.spritecollideany(self.ship, self.aliens):  
        self._ship_hit()
```

Here's the new method `center_ship()`, which belongs in `ship.py`:

```
ship.py def center_ship(self):  
    """Center the ship on the screen."""  
    self.rect.midbottom = self.screen_rect.midbottom  
    self.x = float(self.rect.x)
```

We center the ship the same way we did in `__init__()`. After centering it, we reset the `self.x` attribute, which allows us to track the ship's exact position.

NOTE

Notice that we never make more than one ship; we make only one ship instance for the whole game and recenter it whenever the ship has been hit. The statistic `ships_left` will tell us when the player has run out of ships.

Run the game, shoot a few aliens, and let an alien hit the ship. The game should pause, and a new fleet should appear with the ship centered at the bottom of the screen again.

Aliens That Reach the Bottom of the Screen

If an alien reaches the bottom of the screen, we'll have the game respond the same way it does when an alien hits the ship. To check when this happens, add a new method in `alien_invasion.py`:

```
alien_invasion.py def _check_aliens_bottom(self):  
    """Check if any aliens have reached the bottom of the screen."""  
    for alien in self.aliens.sprites():
```